APM5557: Partial Differential Equations
Weak and Strong Maximum Principles

Both principles say a harmonic function cannot exceed its boundary values. **The weak version says the maximum is reached on the boundary, though it might also be reached inside. The strong version says it cannot be reached inside at all, unless u is constant.**

Throughout, u is harmonic ($\Delta u = 0$) in a bounded open set Ω and continuous on $\overline{\Omega}$. Applying either principle to $-u$ gives the same result for minimums.

The two statements

Weak maximum principle:

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$$

Strong maximum principle: if Ω is connected and u reaches its maximum at an interior point, then u is constant.

Comparison

	Weak	**Strong**
Claim	The maximum occurs on $\partial\Omega$	The maximum occurs only on $\partial\Omega$, unless u is constant
Interior maximum	Not ruled out	Forces u to be constant
Needs Ω connected?	No	Yes
Usual proof	Perturbation and calculus	Mean value property
Typical use	Uniqueness, stability, comparing solutions	Strict inequalities such as $u > 0$ inside

The strong version implies the weak one. If the maximum is reached inside, the component of Ω containing that point has constant u, so the same value is also reached on the boundary.

Why each is true

Weak: let $v = u + \varepsilon|x|^2$, so $\Delta v = 2n\varepsilon > 0$. At an interior maximum of v, all second derivatives would be ≤ 0, so $\Delta v \leq 0$, which is a contradiction. So v reaches its maximum on the boundary, and letting $\varepsilon \to 0$ gives the result for u.

Strong: suppose u reaches its maximum M at an interior point. By the mean value property, M is the average of u over a small ball around that point. Since $u \leq M$ everywhere, that average can only equal M if $u = M$ on the whole ball. So the set where $u = M$ is open. It is also closed in Ω by continuity, so connectedness makes it all of Ω.

Why the hypotheses matter

- **Connectedness:** let Ω be two separate disks, with $u = 1$ on one and $u = 0$ on the other. u is harmonic and reaches its maximum inside, but is not constant. The weak principle still holds, but the strong one fails.
- **Boundedness:** let Ω be the upper half-plane with $u = y$. Then $u = 0$ on the boundary, but u grows without limit inside, so even the weak principle fails.

What each gives you

- **Weak:** if u_1 and u_2 solve the same Dirichlet problem, then $w = u_1 - u_2$ has zero boundary values. Its maximum and minimum are both 0, so $w = 0$. This gives uniqueness.
- **Strong:** suppose Ω is connected, $u \geq 0$ on $\partial\Omega$, and u is not identically zero there. The weak principle gives only $u \geq 0$ inside, but the strong principle gives $u > 0$ everywhere inside. If u were 0 at an interior point, that would be an interior minimum, which would make u constant.